SUPERTEAM IRELAND SUBMISSION

MinKYC

IDEATHON @ DOGPATCH LABS
FRIDAY MARCH 13, 2026

Direct Reference Links:

Website: https://getoutofthatgarden.github.io/minkyc-website/

Repository: https://github.com/GetOutOfThatGarden/MinKYC

Hackathon Submission Note

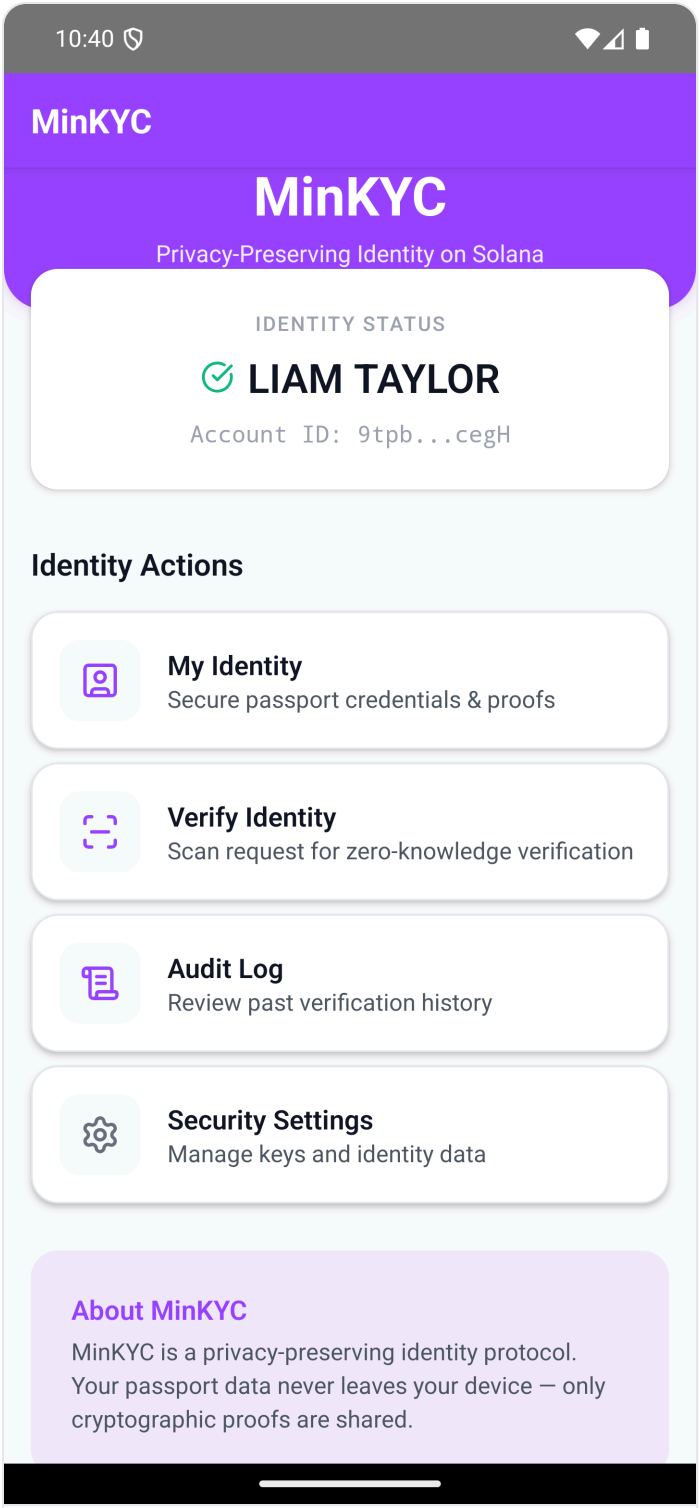
This is my official submission for the Superteam Ireland Ideathon. While I cannot be physically at Dogpatch Labs today, I am actively participating in this guided ideation session to build a concrete foundation for the **Solana Colosseum Hackathon** starting in April. I am grateful for any feedback offered, as it will directly assist in refining the project for the global competition.

> **MinKYC is the privacy-first compliance layer for Solana Mobile. I eliminate KYC data breach liability by proving identity with on-device Zero-Knowledge cryptography—designed specifically for the Solana Seeker.**

In the Web3 ecosystem, users demand privacy, yet regulatory pressures force platforms to collect vast amounts of sensitive personal data. I am solving this paradox by turning the smartphone into a localized compliance hub. Users verify their identity entirely on their device, sharing only mathematical proofs—never their underlying data.

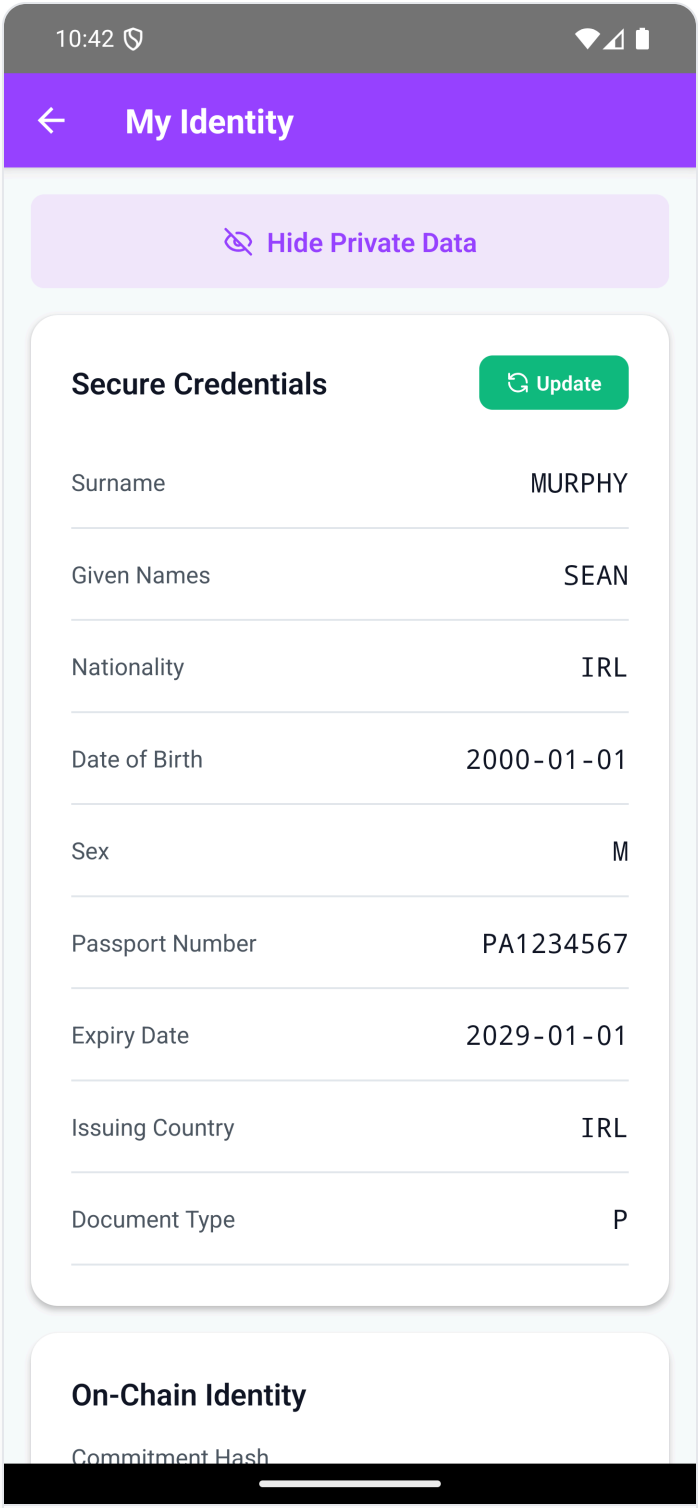
Product UX & Mobile Experience

MinKYC provides a polished, intuitive interface designed to make complex cryptography invisible to the end user.



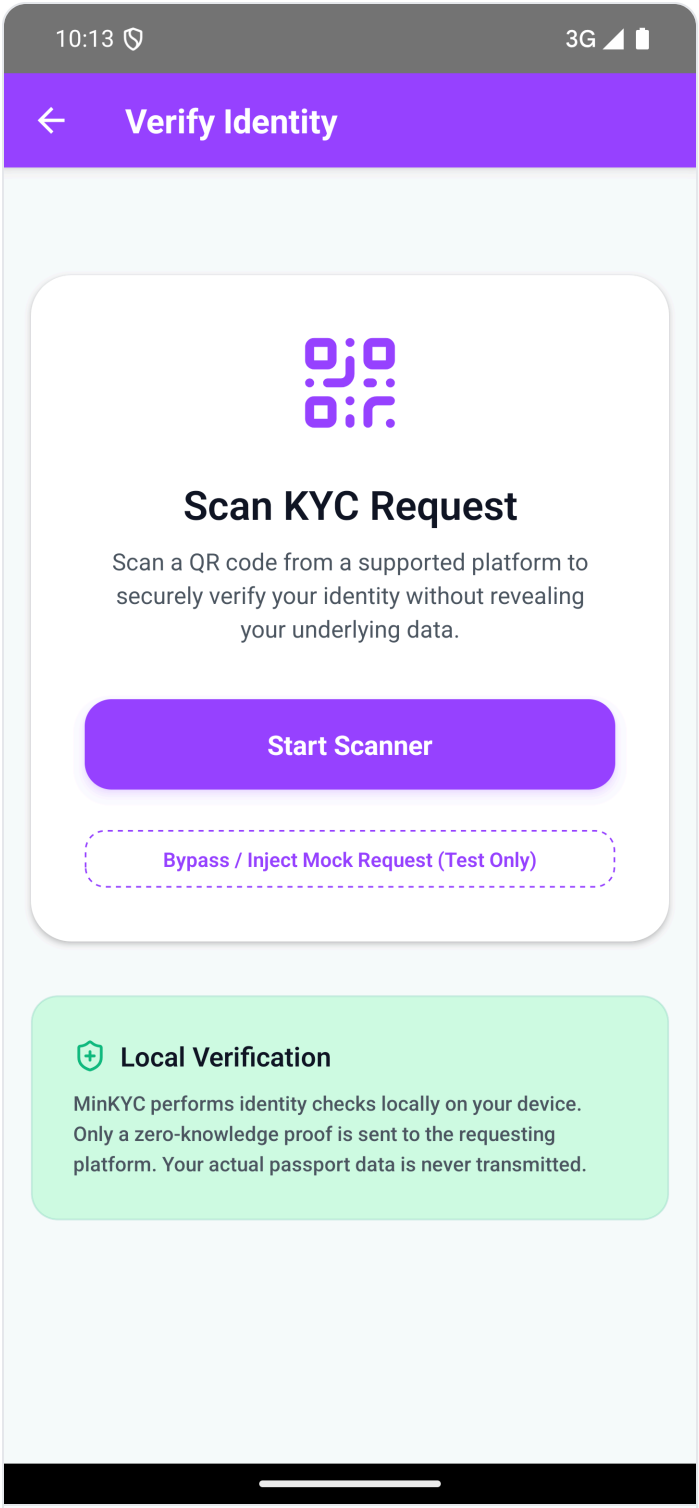
Native Home Interface

A clean dashboard displaying current identity status and quick actions, built for the Seeker screen.



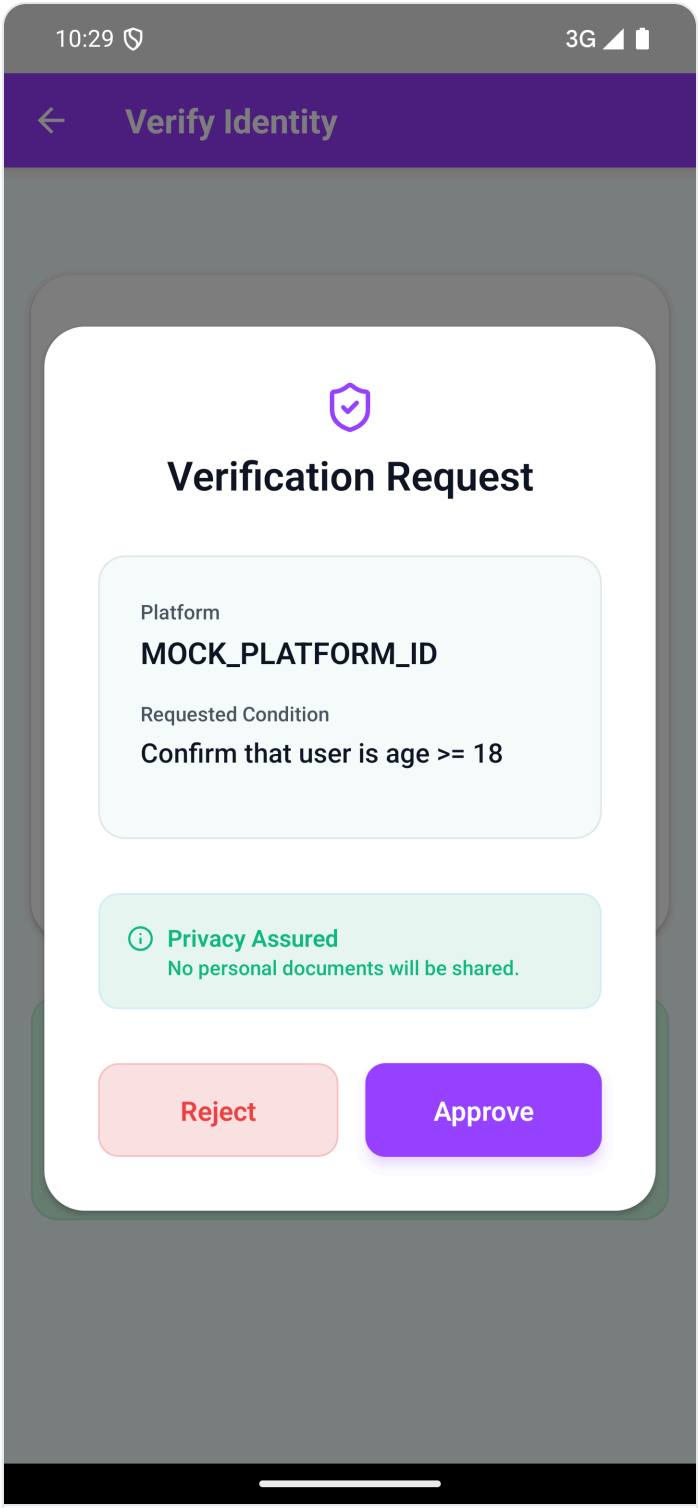
Encrypted Identity Vault

User credentials are encrypted and stored locally. Data never leaves the device.



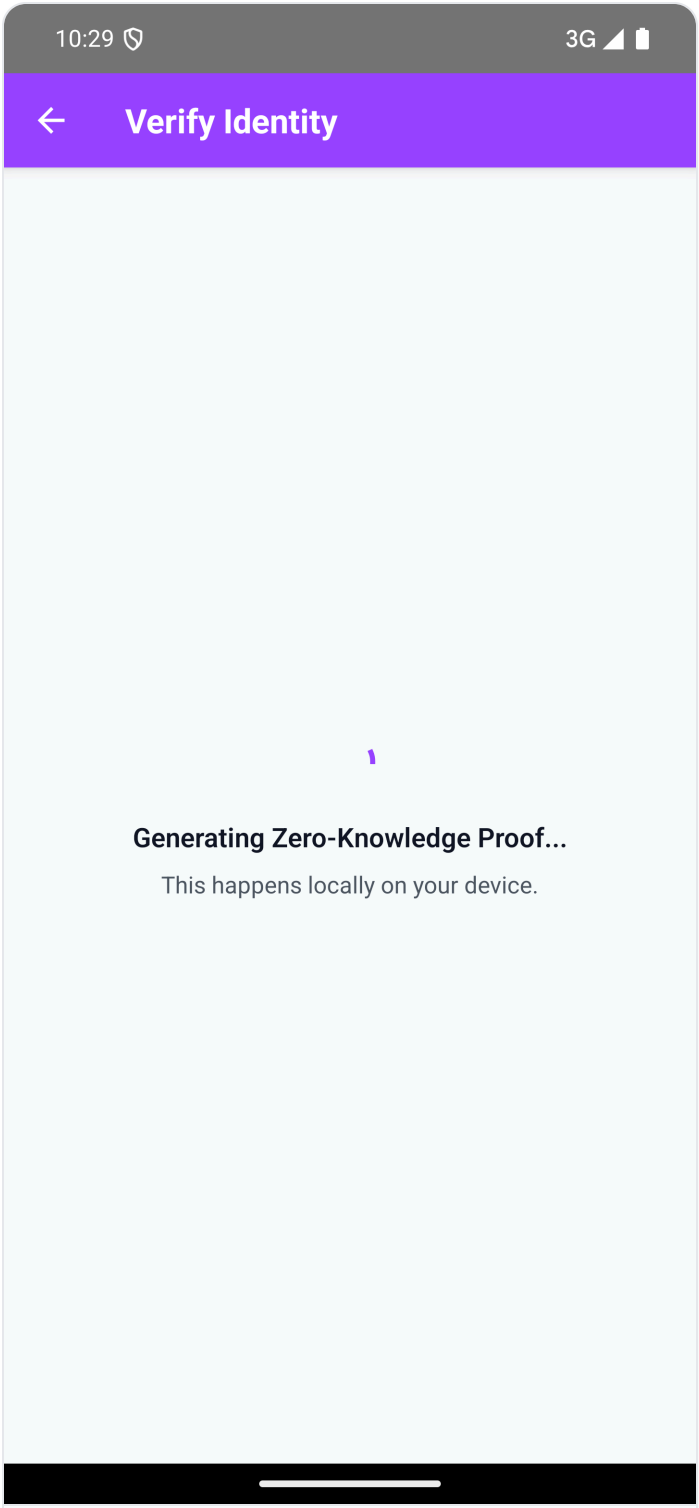
Platform Bridge Scanner

Users scan a QR from a dApp to initiate a seamless mobile-to-web verification handshake.



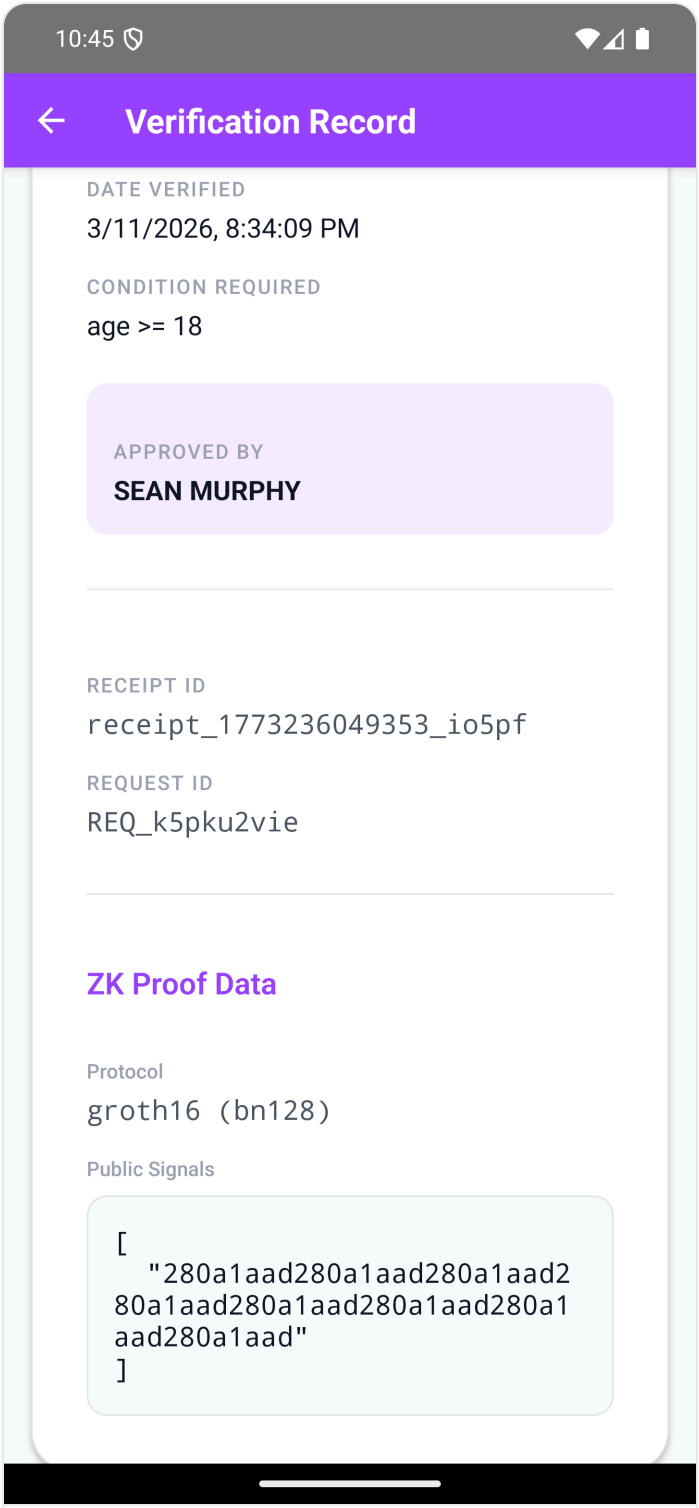
The ZK Proof Request

The user is prompted to generate a specific proof (e.g., Age > 18) without revealing their birthdate.



Instant Verification

Sub-second proof generation and Solana network verification result in a frictionless onboarding.



Immutable Audit Log

A local, private ledger of all successful proofs provided to platforms.

The "Toxic Asset" Problem

For Users: The Identity Honeypot

Centralized KYC databases act as massive honeypots for hackers. When breached, users suffer catastrophic identity theft that cannot be undone.

For Platforms: Crippling Liability

Holding user data is a **Toxic Liability**. The costs of secure storage and regulatory compliance (GDPR/CCPA) stifle early-stage innovation.

The Solution: Zero-Knowledge KYC

I am moving the verification process to the edge—directly onto the user's mobile device—leveraging **Noir ZK circuits**.

- **Local Data Extraction:** NFC passport scanning happens entirely in the device enclave. PII never touches a server.
- **ZK Proof Generation:** The app generates Noir ZK proofs locally (e.g., "Age > 18") with zero data egress.
- **Solana Verification:** Platforms receive a `true/false` confirmation via a cryptographic proof verified on-chain.

Why Solana?

To replace traditional APIs, KYC verifications must be instant and cheap. Solana's high throughput and sub-dollar fees allow for sub-second ZK proof verification and the minting of immutable audit receipts, creating a transparent yet private compliance trail.

Technical Stack

- Solana Mobile Stack (SMS)
- Noir (ZK Circuits)
- React Native
- NFC Data Protocol
- Rust

© 2026 MinKYC — Superteam Ireland Ideathon Project